

Assignment of Chapter Two (A)

1. Fill-in-the-blank Questions:

1. The Logic Algebra contains _____, _____, and _____, such three primitive functions/operators.
2. The three important rules in the logic algebra refer to _____, _____ and _____.
3. In the logic algebra, if $A \cdot B = A + B$, then, $A=B$ is _____ (True/False).

2. Short Answer Questions : Prove the following logic expression equations

1.
$$\overline{A}B + \overline{A}C + B\overline{C} = \overline{A}B\overline{C}(A + B + C)$$

2.
$$AB\overline{C} + \overline{A}BC + AB = \overline{A}BC$$

3.
$$AB + \overline{A}C + \overline{B}C = AB + C$$

$$4. \quad BC + D + \bar{D}(\bar{B} + \bar{C})(DA + B) = B + D$$

$$5. \quad (A + B)(A + \bar{B})(\bar{A} + B)(\bar{A} + \bar{B}) = 0$$

$$6. \quad AB + BC + AC = (A + B)(B + C)(A + C)$$

$$7. \quad \overline{(\bar{A} + \bar{B} + \bar{C})(\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}) + \bar{A}\bar{B}\bar{C}} = (A + B + C)(AB + BC + AC) + ABC$$

8. Find the **complement and dual forms** of the following functions

(1) $F = AB + \bar{A} \bar{B}$

(2) $F = (A + B)(\bar{A} + C)(C + DE) + \bar{E}$

(3) $F = \overline{[\bar{A}B + (C + D)\bar{A}C]}$

(4) $F = A[\bar{B} + (C\bar{D} + \bar{E})G]$